# Karthika Pillai Kumar

**Data Scientist & Engineer** | Melbourne, Australia

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## PROFILE

Master of Data Science student at Monash (2025-2026), with software engineering experience on IBM watsonx.data (Presto). Open to graduate and junior data roles in Melbourne. Focus: predictive modelling, reliable data pipelines and applied AI.

## EXPERIENCE

**Software Developer | IBM India Software Labs, India**
August 2022 - January 2025

Worked on Presto within watsonx.data across security, performance monitoring and developer tooling. Contributed audit logging, automated security checks and caching improvements. Built Python tools to streamline quality assurance.

**Coding Instructor | Junior Engineers, Melbourne**
August 2025 - Present

Teach Python, introductory data science and AI concepts. Support instructor onboarding and explain complex concepts through practical lessons.

## SELECTED PROJECTS

**Voltwise | Python, scikit-learn, SQL, dbt, DuckDB, FastAPI, Ollama**

Forecast battery capacity ten discharge cycles ahead using NASA laboratory data. Leave-one-battery-out evaluation across four cells: 0.024 Ah MAE across 560 forecast windows, 44.7% lower error than persistence. Includes 13 dbt checks and a local, read-only AI agent. Research prototype; not validated on vehicle fleets. Source code

**Nexora (DataMedic) | Data pipeline diagnostics**

Interactive demonstration of pipeline diagnosis and recovery using synthetic scenarios and recorded results. Makes investigation steps visible. Source code

**Sensory-Aware Route Planner | Python, OpenStreetMap, GitHub Actions**

Melbourne walking-route explorer using street and address data plus timestamped pedestrian-counter snapshots. Validation flags stale or incomplete observations; unavailable noise and light measurements remain unknown. Source code

## EDUCATION

**Master of Data Science** | Monash University, Melbourne | 2025-2026
**B.Tech (Honours), Computer Science Engineering** | TKM College of Engineering, India

## SKILLS

**Data and modelling:** Python, SQL, NumPy, scikit-learn, model validation.
**Analytics and applications:** dbt, DuckDB, semantic metrics, FastAPI, local Ollama agents.
**Professional platform work:** Presto, Java, audit logging, performance monitoring, Python tooling.